

NT14

$$\xi \sim N(a, 2)$$

$$\eta \sim N(b, 1)$$

$$\vec{x}_n = \{-1, 11; -6, 10; 2, 42\}$$

$$\vec{y}_m = \{-2, 29; -2, 91\}$$

$$\alpha = 0,05$$

$$H_0: a = b; H_1: a > b$$

ξ:

$$\frac{\bar{x} - a}{\sigma_x} \sqrt{n} \sim N(0, 1)$$

$$\bar{x} - a \sim N\left(0, \frac{\sigma_x^2}{n}\right)$$

η:

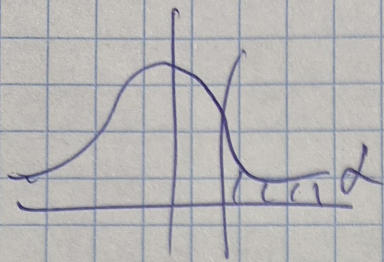
$$\frac{\bar{y} - b}{\sigma_y} \sqrt{m} \sim N(0, 1)$$

$$\bar{y} - b \sim N\left(0, \frac{\sigma_y^2}{m}\right)$$

$$\bar{x} - a - (\bar{y} - b) \sim N\left(0, \frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}\right) = N\left(0, \frac{7}{6}\right)$$

$$\Delta = \frac{\bar{x} - \bar{y} - (a - b)}{\sqrt{7/6}} \sim N(0, 1)$$

$$H_0 \rightarrow \tilde{\Delta} = \frac{\bar{x} - \bar{y}}{\sqrt{\frac{7}{6}}} \sim N(0, 1)$$



$$G: \tilde{\Delta} \geq U_{1-\alpha}$$

$$U_{1-\alpha} = 1,645$$

$$\Rightarrow G: \tilde{\Delta} \geq 1,645$$

Данные выданы: $\bar{x} = 7,597$ $\bar{y} = -2,8$

$$\Rightarrow \tilde{\Delta} = 0,929 < 1,645$$

$\Rightarrow$ Нет оснований отвергать H_0

$$p\text{-value} = P(\Delta \geq \tilde{\Delta} | H_0) = 1 - \Phi(0,929) = 0,1762 > 0,05$$

$\Rightarrow$ Нет оснований отвергнуть H_0

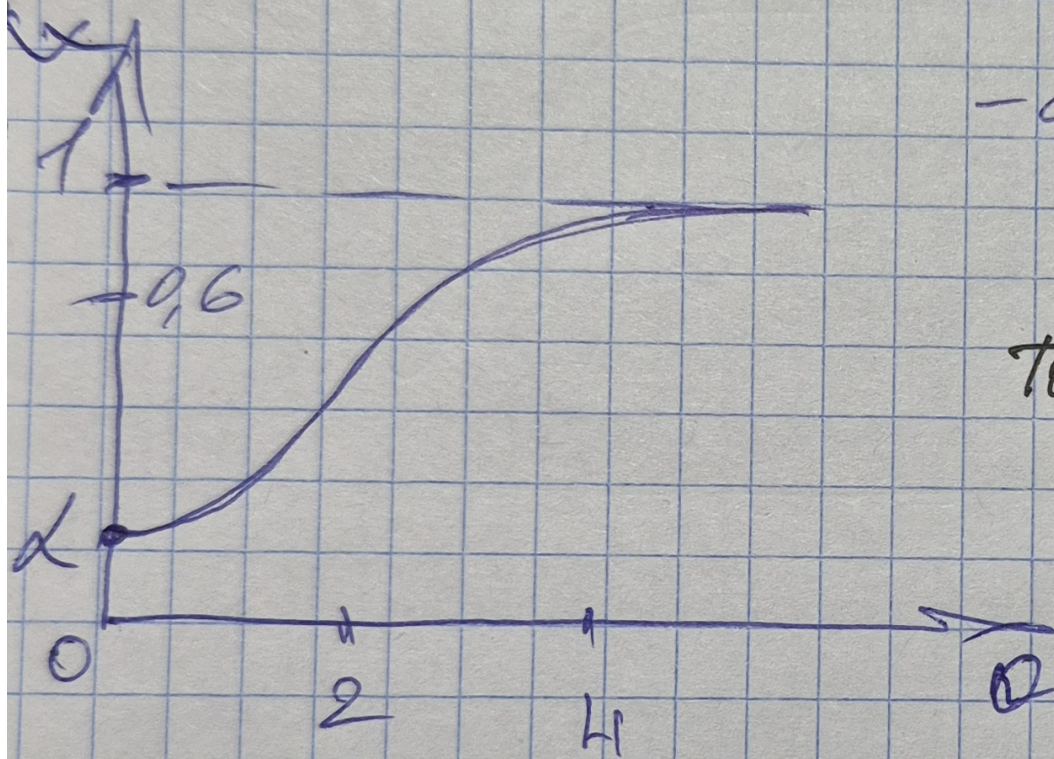
$$W = P(\tilde{\Delta} \geq 1,645 | H_1) = P\left(\frac{\bar{x} - \bar{y}}{\sqrt{\frac{7}{6}}} \geq 1,645 \mid H_1\right)$$

$$= P\left(\frac{\bar{x} - \bar{y} - 0}{\sqrt{\frac{7}{6}}} \geq 1,645 - \frac{0}{\sqrt{\frac{7}{6}}} \mid H_1\right) =$$

$$= 1 - \Phi\left(1,645 - \frac{0}{\sqrt{\frac{7}{6}}}\right), \text{ где } 0 = a - b$$

$$1,645 - \frac{0,7}{\sqrt{6}}$$

$$\Rightarrow W(t) = 1 - \int_{-\infty}^{\frac{1,645 - \frac{0,7}{\sqrt{6}}}{\sqrt{24}}} e^{-\frac{x^2}{2}} dx$$



Расчёт и
точный график $W(t)$
лежат в файле
task_14.ipynb